Results Checkpoint: PPE Safety Equipment Detection

P14 — Computer Vision 2026 — 2 November 2026

1. Completed experimental evaluation

We compare COCO-pretrained YOLO11n and YOLO11s on all 11 Construction-PPE classes. The frozen split contains 1,265 training, 73 validation and **78 test images (924 boxes)**. Both models were fine-tuned for 50 epochs, batch 16, 640-pixel inputs, SGD learning rate 0.01, momentum 0.937, weight decay 0.0005, three warm-up epochs and cosine decay. Training uses flips and mild scale/translation/HSV augmentation, without mosaic, mixup or copy-paste. We report all seeds 42, 43 and 44, selecting each checkpoint by validation mAP50:95.

Stage 4 selected confidence **0.20 for YOLO11n and 0.30 for YOLO11s** by mean validation macro-F1. These values and the six checkpoint hashes were frozen before test inference. There is no test threshold sweep or test-based seed selection. Both models use square letterboxing to 640×640, FP32, class-aware NMS IoU 0.70 and at most 300 detections. AP uses confidence 0.001, library multi-label expansion and IoU thresholds 0.50:0.05:0.95. Fixed-threshold metrics use standard single-label predictions and greedy, same-class, one-to-one matching in descending IoU order at IoU ≥ 0.50. Macro-F1 is the mean of the 11 class F1 scores, $F1_c = 2TP_c/(2TP_c + FP_c + FN_c)$; macro P/R average class P/R. All test classes have ground truth.

Test metric	YOLO11n	YOLO11s
mAP@0.50:0.95	0.1874 ± 0.0037	0.1902 ± 0.0067
mAP@0.50	0.4413 ± 0.0150	0.4505 ± 0.0203
Macro precision	0.4731 ± 0.0547	0.5512 ± 0.0307
Macro recall	0.4632 ± 0.0061	0.4525 ± 0.0164
Macro-F1	0.4547 ± 0.0238	0.4835 ± 0.0218
Macro-F1 (reference confidence 0.25)	0.4593 ± 0.0211	0.4791 ± 0.0218

All values are **mean ± sample SD across three seeds**, not confidence intervals. P/R/F1 use each model's frozen threshold; AP is a ranked-prediction metric. YOLO11s gains 0.0028 absolute mAP50:95 and 0.0288 macro-F1, with higher precision but slightly lower recall. The AP gain is small relative to seed variation and does not establish statistical significance. YOLO11n's reference F1 at 0.25 exceeds its selected-threshold test F1; we retain 0.20 to respect validation-only selection.

Model	Seed	mAP50:95	mAP50	Macro P	Macro R	Macro-F1
yolo11n	42	0.1905	0.4547	0.5056	0.4625	0.4739
yolo11n	43	0.1833	0.4251	0.4100	0.4574	0.4280
yolo11n	44	0.1885	0.4441	0.5039	0.4695	0.4621
yolo11s	42	0.1825	0.4279	0.5275	0.4336	0.4594
yolo11s	43	0.1933	0.4565	0.5403	0.4620	0.4890
yolo11s	44	0.1947	0.4672	0.5859	0.4620	0.5020

2. Runtime and computational cost

Model	Params (M)	Weights (MB)	CPU ms	IQR (ms)	Train min/seed
yolo11n	2.61	5.51	50.21	46.26–53.38	50.7 ± 0.9
yolo11s	9.43	19.18	79.80	68.08–85.78	88.9 ± 0.1

New controlled CPU benchmark: Apple M2 Max (Ultralytics CPU identification), four PyTorch threads, batch one, FP32, seed 42, confidence 0.25; 20 warm-up and 100 timed calls per model on the same ordered, in-memory test images. Median and interquartile range (IQR) include preprocessing, forward inference, NMS and predict-call overhead; disk reading, loading weights and plotting are excluded. YOLO11s costs 1.59 times the median latency. Parameter counts are from the unfused checkpoints; MB means 10^6 bytes.

Training wall times used Apple M4/MPS for YOLO11n and M2 Max/MPS for YOLO11s (three concurrent jobs, CPU loss/NMS workaround). They are **not a controlled training-speed comparison**. Historical Stage 4 MPS medians (7.58/10.03 ms) are separate from these CPU results. Test software: Python 3.14.7, PyTorch 2.14.1, Ultralytics 8.4.175, macOS 15.7.2 arm64.

3. Class performance and difficult subsets

Class	Train	Test	AP50 (n)	AP50 (s)	Recall (n)	Recall (s)
helmet	1454	161	0.903 ± 0.005	0.904 ± 0.008	0.882 ± 0.006	0.878 ± 0.013
gloves	1351	57	0.322 ± 0.018	0.351 ± 0.023	0.357 ± 0.027	0.322 ± 0.037
vest	1376	150	0.830 ± 0.015	0.812 ± 0.027	0.833 ± 0.013	0.813 ± 0.040
boots	1413	144	0.339 ± 0.023	0.352 ± 0.014	0.412 ± 0.083	0.377 ± 0.021
goggles	501	9	0.106 ± 0.065	0.202 ± 0.115	0.037 ± 0.064	0.111 ± 0.111
none	677	51	0.558 ± 0.085	0.558 ± 0.009	0.673 ± 0.011	0.647 ± 0.020
Person	1884	198	0.768 ± 0.032	0.782 ± 0.017	0.790 ± 0.026	0.783 ± 0.013
no_helmet	400	40	0.377 ± 0.051	0.312 ± 0.044	0.442 ± 0.095	0.375 ± 0.025
no_goggle	337	33	0.382 ± 0.086	0.486 ± 0.080	0.424 ± 0.061	0.465 ± 0.063
no_gloves	442	58	0.157 ± 0.040	0.153 ± 0.006	0.172 ± 0.034	0.178 ± 0.010
no_boots	88	23	0.111 ± 0.011	0.043 ± 0.019	0.072 ± 0.066	0.029 ± 0.025

Counts are ground-truth boxes; AP50 and recall are mean ± SD across seeds. Full AP50:95, precision, F1 and TP/FP/FN counts are exported per class and seed. Helmet, vest and Person are strongest. Goggles (only nine test boxes) and missing-equipment classes remain difficult; `no_boots` AP50 is lower for YOLO11s. Thus increasing model size does not improve every class. Class imbalance is relevant, but object scale, ambiguous source boxes and domain diversity also affect performance; these data do not isolate a causal imbalance effect.

Object subset	Count	Recall (n)	Recall (s)
size: small	98	0.354 ± 0.026	0.310 ± 0.012
size: medium	469	0.548 ± 0.013	0.527 ± 0.018
size: large	357	0.810 ± 0.018	0.810 ± 0.010
visibility: clear	259	0.835 ± 0.010	0.829 ± 0.022
visibility: occluded	72	0.676 ± 0.021	0.676 ± 0.069
visibility: uncertain	190	0.432 ± 0.045	0.388 ± 0.008
truncated: yes	24	0.819 ± 0.024	0.833 ± 0.042
truncated: no	309	0.796 ± 0.006	0.790 ± 0.034
truncated: uncertain	188	0.433 ± 0.048	0.388 ± 0.011

Size: all 924 objects are partitioned using box area after aspect-preserving resize to 640 pixels: small $A < 32^2$, medium $32^2 \leq A < 96^2$, large $A \geq 96^2$. Matching is done on the full image first; subgroup recall then counts matched targets, so a prediction cannot be reused across groups. This is object-weighted recall, not macro recall. Class-by-size results appear in Section 5. Both models miss most small objects, and YOLO11s has lower small-object recall at its frozen operating point. These thresholds differ, so the comparison combines model and deployment-threshold effects.

Visibility and truncation: these rows cover only the five PPE classes (521 objects). The primary visibility comparison uses 259 clear and 72 occluded objects; 190 uncertain objects are excluded from that comparison and shown separately for transparency. Occluded recall is 0.676 for both models, versus approximately 0.83 for clear PPE. These are observational associations, not an isolated occlusion effect. Size distributions differ:

PPE visibility	Small	Medium	Large
clear	13	175	71
occluded	1	33	38
uncertain	49	135	6

The groups also differ in class composition: 58 of 72 occluded objects are vests; occluded goggles have no support. The truncation rows are a separate partition (24 yes, 309 no, 188 uncertain); higher recall for the small truncated group does not imply truncation helps detection. **Review limitation:** visibility/truncation labels come from one assistant visual review before predictions, not two independent human reviewers; the planned Member 4 review has not occurred. Class-specific counts and recalls, including sparse groups, are provided in the CSV outputs.

Visual comparison of test performance

The plots summarize the same measurements as Section 3; tables retain the exact values. Error bars show **sample standard deviation across seeds 42, 43 and 44**, not confidence intervals. Counts n refer to ground-truth objects in the fixed test split.

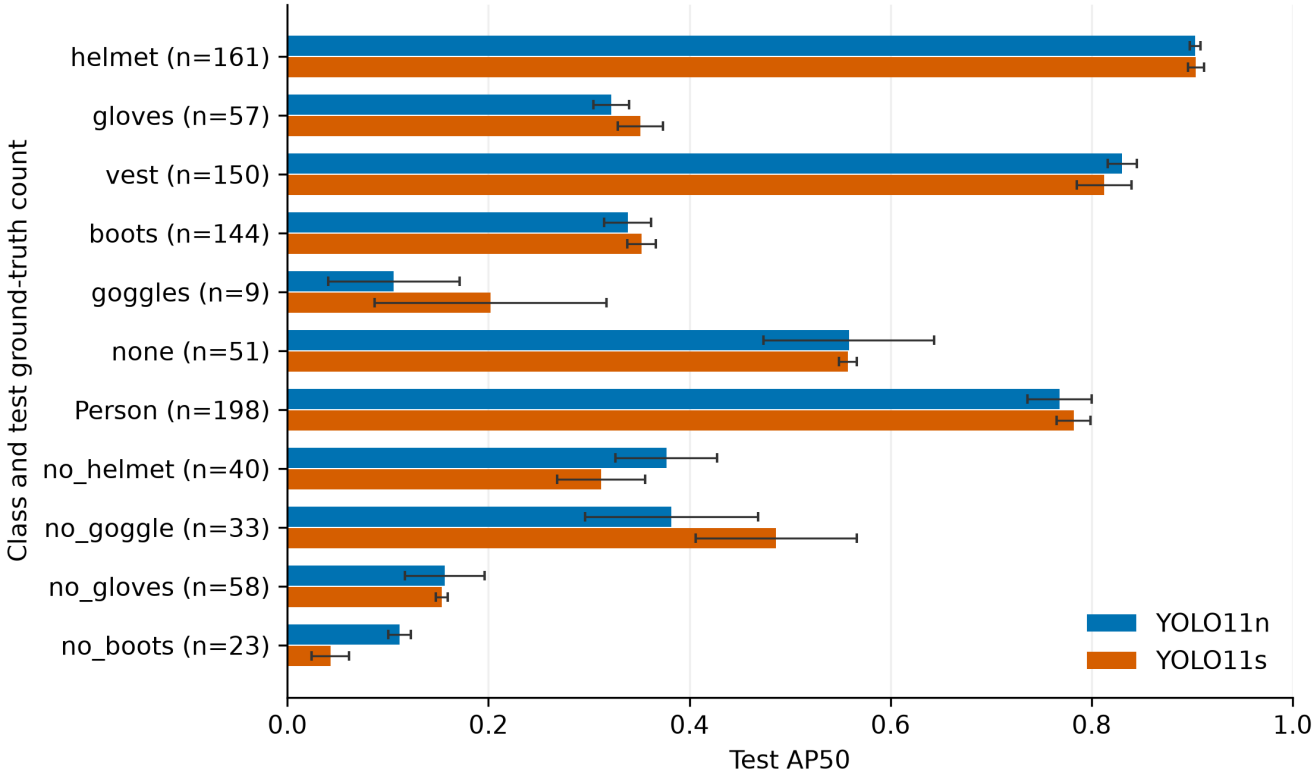


Figure 1. Per-class AP50. The larger model improves some classes, especially `goggles` and `no_goggle`, but loses performance on `no_helmet` and `no_boots`. The high variability and nine-object support for goggles limit conclusions. AP uses ranked predictions at confidence 0.001, independently of the fixed deployment thresholds.

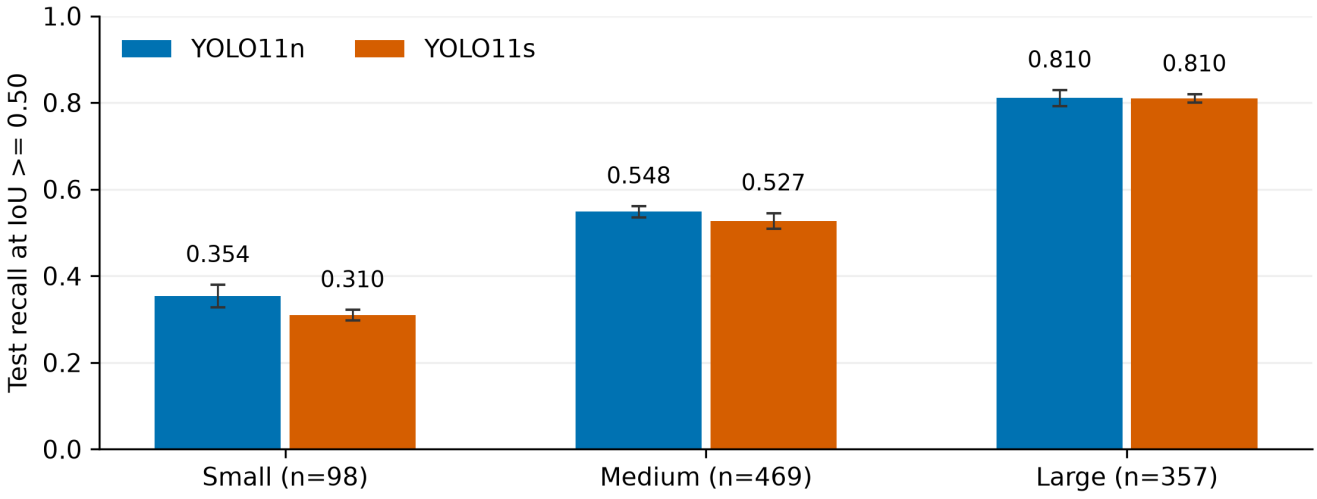
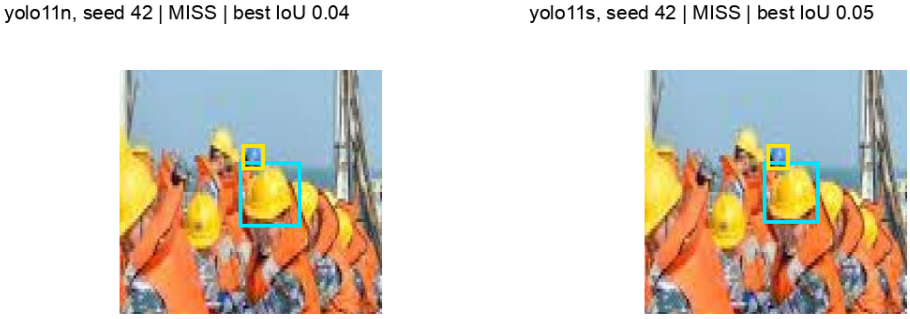


Figure 2. Recall by object size. Small objects have the lowest recall for both models. Recall uses IoU ≥ 0.50 and the frozen confidence thresholds: 0.20 for YOLO11n and 0.30 for YOLO11s. The comparison therefore includes both model and operating-point effects; counts and class composition differ across sizes. All 924 test objects are included, with size boundaries 32^2 and 96^2 pixels after resize.

4. Three representative observed failures

Both columns use seed 42 and the frozen model thresholds. **Yellow** marks the selected ground-truth object; **cyan** marks the retained same-class prediction with the highest IoU to that object, when overlap is nonzero. Crops are enlarged for readability; inference used the full image. Examples are illustrative, not estimates of error frequencies. Source image names and original one-based label lines below permit exact recovery.



Small-object miss — `images/test/image182.jpg, line 7, helmet.` The target occupies approximately 24×28 pixels at 640 resolution ($A = 661$ px^2). Neither model matches it. The cyan boxes belong to a larger neighbouring helmet and are already matched to another target (IoU to this target 0.045/0.050). Limited pixel detail and a crowded, visibly transformed image plausibly explain the miss. Its visibility tag is uncertain; this example is not evidence for an occlusion effect.



Occlusion-associated miss — `images/test/image1037.jpeg, line 10, vest.` This medium target has $A = 4,671$ px^2 and is tagged occluded. The yellow region is partly hidden among overlapping workers and arms. Both models detect the neighbouring worker's larger vest (cyan), already assigned to another target, but not this box (best IoU 0.185/0.159). Occlusion and difficult instance separation are plausible causes. The source vest boundary is also ambiguous, so annotation quality may contribute; this single case does not establish causality.



Localization false positive — `images/test/image1125.jpg, line 10, Person.` Both models emit a broad Person box spanning multiple overlapping people under umbrellas. Confidence is 0.837/0.798, yet IoU to the selected person is only 0.387/0.330. Each cyan prediction is unmatched to *any* target at IoU 0.50, making it a false positive; the selected person is also a false negative. Overlap, low contrast and merged silhouettes plausibly cause the localization error. Raising confidence alone would retain these high-confidence errors.

5. Class breakdown of size-dependent recall

Class	Size	Count	Recall (n)	Recall (s)
helmet	small	33	0.758 ± 0.030	0.727 ± 0.052
helmet	medium	111	0.907 ± 0.005	0.907 ± 0.014
helmet	large	17	0.961 ± 0.068	0.980 ± 0.034
gloves	small	14	0.167 ± 0.041	0.095 ± 0.041
gloves	medium	41	0.439 ± 0.049	0.415 ± 0.042
gloves	large	2	0.000 ± 0.000	0.000 ± 0.000
vest	medium	54	0.809 ± 0.028	0.741 ± 0.085
vest	large	96	0.847 ± 0.022	0.854 ± 0.018
boots	small	16	0.354 ± 0.253	0.188 ± 0.125
boots	medium	128	0.419 ± 0.063	0.401 ± 0.009
goggles	medium	9	0.037 ± 0.064	0.111 ± 0.111
none	small	3	0.000 ± 0.000	0.000 ± 0.000
none	medium	11	0.515 ± 0.139	0.424 ± 0.052
none	large	37	0.775 ± 0.041	0.766 ± 0.016
Person	small	3	0.000 ± 0.000	0.000 ± 0.000
Person	medium	3	0.444 ± 0.192	0.333 ± 0.000
Person	large	192	0.807 ± 0.024	0.802 ± 0.014
no_helmet	small	3	0.000 ± 0.000	0.000 ± 0.000
no_helmet	medium	26	0.385 ± 0.133	0.308 ± 0.067
no_helmet	large	11	0.697 ± 0.052	0.636 ± 0.091
no_goggle	small	4	0.333 ± 0.144	0.417 ± 0.144
no_goggle	medium	29	0.437 ± 0.072	0.471 ± 0.080
no_gloves	small	12	0.028 ± 0.048	0.028 ± 0.048
no_gloves	medium	44	0.212 ± 0.026	0.205 ± 0.023
no_gloves	large	2	0.167 ± 0.289	0.500 ± 0.000
no_boots	small	10	0.000 ± 0.000	0.000 ± 0.000
no_boots	medium	13	0.128 ± 0.118	0.051 ± 0.044

Support is the number of test boxes in each class/size cell; recall is mean ± SD over the three seeds. Only nonempty cells are shown. Aggregate size recall must be read together with these counts. Goggles and several missing-equipment cells have very limited support.

6. Interpretation and reproducibility

YOLO11s improves mean test macro-F1, but provides only a small AP gain, costs about 3.6 times as many parameters and 1.59 times the CPU latency, and does not solve small-object or missing-equipment failures. YOLO11n remains a reasonable efficiency baseline; YOLO11s is preferable when the measured precision/F1 gain matters more than latency. These observations apply to this fixed test set and operating points. Three seeds measure training variability on one split; they do not measure uncertainty over new sites. Dataset labels identify equipment categories without assigning PPE to individual workers.

The report uses the retained reproduction checkpoints. The lost Stage 3 T4 experiment is historical and is not included in test aggregates. The test split was used only after Stage 4 selection; future changes should be tuned on validation or new development data. No additional training was needed for this checkpoint.

Reproduce from the repository root:

```
.venv/bin/python scripts/evaluate_test.py --device cpu --benchmark
.venv/bin/python scripts/build_results_report.py
```

In `planning/latex`, compile `results_checkpoint.tex` with Tectonic or run pdfLaTeX twice. Preparation and output details are documented in the Results checkpoint README. Checkpoint, test-content, annotation and validation-summary SHA-256 values are retained in JSON provenance records. Caches require matching content and environment fingerprints.

Evidence: the test summary, per-seed/per-class/subgroup CSVs and `failure_examples.json` in `reports/results_checkpoint`; Stage 4 validation summary; retained training logs; dataset split manifest and visibility review. Source attribution and archive checksum are in `data/ATTRIBUTION.md` and `data/source.json`. Repository: https://github.com/Ta6leTkaa/intro-to-CV-Project.